CARECONNECT EHR SYSTEM

Project Code: EHRS01

Official Academic Project Synopsis & System Specification Document

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Program: B.Tech / M.Tech Computer Science & Engineering Major Project
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GitHub Repository: https://github.com/scrollingreel/P_003-CareConnect-Patient-EHR-HCL-TECH-PROJECT

This document provides the formal 25-section project synopsis for CareConnect, detailing system architecture, functional requirements, HCL epics, database design, and verified evaluation outcomes.

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1 & 2 & 3 & 4. Project Title, Code & Student Information

PROJECT TITLE: CareConnect – Patient-Provider Electronic Health Record System

PROJECT CODE: EHRS01

DOMAIN: Healthcare and Pharma

PRACTICE: Academic Center of Excellence (COE) / Healthcare Practice

DELIVERY METHODOLOGY: Agile / Scrum (4 Sprints)

STUDENT / CANDIDATE INFORMATION PLACEHOLDERS:

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• Degree / Specialization: B.Tech / M.Tech in Computer Science & Engineering

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• Official GitHub Repository: https://github.com/scrollingreel/P_003-CareConnect-Patient-EHR-HCL-TECH-PROJECT

5. Abstract

Modern hospital ecosystems frequently suffer from fragmented patient documentation, manual paper-based charting, siloed diagnostic information systems, and lack of computerized medication safety checks. CareConnect (Project Code: EHRS01) is an enterprise-grade Electronic Health Record (EHR) and clinical workflow orchestration system engineered to solve these operational bottlenecks. Built upon a modern decoupled architecture comprising a Spring Boot 3.5.7 (Java 24) RESTful backend and a React 18 Single Page Application, CareConnect unifies patient registration, multi-specialty clinical documentation (SOAP, Cardiology, Discharge summaries), Computerized Provider Order Entry (CPOE) for laboratory, imaging, and procedures, electronic prescribing with automated Drug-Drug Interaction (DDI) and allergy contraindication screening, diagnostic results integration with real-time critical panic alerts, and an engaging patient self-service portal. The system introduces tamper-resistant encounter note finalization (enforcing HTTP 409 Conflict on unauthorized modifications of signed records), interoperability bridges conforming to HL7 v2.5.1 and HL7 FHIR Release 4, and an auditable compliance log. Comprehensive automated verification confirms 100% test success across all mandatory HCL project epics.

6. Introduction

Healthcare delivery involves coordinated collaboration across diverse clinical specialties, nursing teams, laboratory technicians, radiologists, administrative personnel, and patients. Traditional hospital record systems often maintain separated silos for patient demographics, inpatient clinical charting, laboratory information systems (LIS), picture

archiving systems (PACS), and pharmacy records. These fragmented silos cause transcription errors, delays in critical diagnostic response, adverse drug events, and patient disengagement.

CareConnect was conceptualized and developed under the HCL Project Olympus framework as an end-to-end, unified EHR platform that unifies all clinical, diagnostic, medication, and communication workflows into an integrated operational fabric. By enforcing strict Role-Based Access Control (RBAC), ACID-compliant transactional consistency across related healthcare modules, and transparent diagnostic interoperability adapters, CareConnect delivers a high-reliability software foundation ready for clinical COE evaluation.

7. Problem Statement

Contemporary healthcare provider organizations face four acute systemic challenges:

1. Fragmented Patient Records: Patient history, chronic ailments, allergies, and diagnostic results are scattered across disparate databases and legacy software, requiring clinicians to cross-reference multiple tools during acute care encounters.

2. Unreliable and Tamper-Prone Clinical Documentation: Paper records and unversioned digital forms permit silent overwrites, unauthorized post-facto alterations, and lack standardized specialty templates (e.g., SOAP, Cardiology consults).

3. Medication Errors & Adverse Drug Events: Absence of point-of-care pharmacology cross-referencing leads to prescribing contraindicated drug pairs (e.g., ACE inhibitors with potassium-sparing diuretics) and administering medications without synchronized Medication Administration Records (MAR).

4. Delayed Diagnostic Response & Disconnected Patients: Critical laboratory abnormalities (e.g., severe hypokalemia or elevated cardiac troponin) often stall in technician queues without automated physician panic alerting, while patients lack self-service access to released records and care team messaging.

8. Objectives

The primary technical and operational objectives of CareConnect (EHRS01) include:

• Objective 1: Streamline Patient Onboarding & Demographics — Provide unified digital intake capturing demographic information, insurance policy coverage, chronic illness history, and verified allergies.

• Objective 2: Enforce Standardized & Tamper-Evident Documentation — Provide structured templates (SOAP, Cardiology, Discharge) with clear separation of Assessment and Plan, locking finalized notes against silent modification.

• Objective 3: Implement Multi-Departmental CPOE — Enable healthcare providers to order laboratory tests, imaging modalities, and interventional procedures with status monitoring from order dispatch to fulfillment.

• Objective 4: Elevate Medication Safety — Integrate electronic prescribing with real-time RxNorm-aligned Drug-Drug Interaction screening and nurse-verified MAR administration event tracking.

• Objective 5: Diagnostic Integration & Critical Alerting — Centralize lab and imaging result retrieval, automatically dispatching high-priority physician notifications upon detection of critical panic values.

• Objective 6: Standards-Compliant Interoperability — Provide modular gateways formatting clinical observations into HL7 v2.5.1 ORU^R01 messages and HL7 FHIR R4 DiagnosticReport resources.

• Objective 7: Empower Patient Self-Service — Deliver a secure patient portal for viewing released records, monitoring active medications, booking consultations, and conducting secure messaging.

9. Project Scope

The scope of CareConnect encompasses hospital-wide clinical, diagnostic, and patient engagement workflows:

• In-Scope Functions: Outpatient & Inpatient Registration; Clinical Encounter Note Authoring with Digital Signing; CPOE Laboratory, Imaging, and Procedure Ordering; Prescription Authoring with DDI Warnings; Medication Administration Records (MAR); Diagnostic Result Review with Critical Flags; Patient Appointment Booking; Secure Patient-Provider Messaging; Operational Hospital Analytics; HL7/FHIR Data Transformation Adapters.

• Explicit Out-of-Scope / Boundaries: Autonomous AI medical diagnosis or automated prescribing without clinician review; live commercial database licensing (e.g., proprietary First Databank / Medi-Span credentials are appropriately represented by a validated NLM RxNorm demonstration gateway); unauthenticated internet-facing deployment without institutional VPN.

10. Existing System Analysis

Traditional and legacy healthcare software environments exhibit key vulnerabilities:

• Manual Data Re-entry: Clinicians transcribe test results from printed lab slips into separate electronic forms, creating transcription errors.

• Vulnerable Audit Trails: Many legacy relational schemas permit direct SQL updates or unconstrained HTTP PUT mutations to clinical notes without state validation.

• Disjointed Order Life-Cycles: Laboratory orders exist only within LIS systems, leaving physicians unaware of pending vs completed tests until manual paper delivery.

• Neglected Patient Portals: Patient portals, when present, often operate as third-party bolt-ons that lack real-time synchronization with physician charting.

11. Proposed System Architecture

The proposed CareConnect architecture adopts a modern, service-oriented multi-tier paradigm:

• Client Tier (React 18): High-responsiveness single-page application utilizing React Context for state management, TanStack React Query for cached asynchronous queries, Tailwind CSS for accessible clinical design systems, and Lucide React icons.

• Application & Security Tier (Spring Boot 3.5.7): Enterprise Java backend with stateless JWT bearer authentication, Spring Security filter chains, BCrypt password hashing, and transactional service boundaries (@Transactional).

• Clinical Pharmacology Safety Engine: Standalone service implementing standardized pharmacology logic to evaluate drug-drug contraindications and allergy cross-hypersensitivities.

• Interoperability Gateway: Modular transformer emitting HL7 v2.5.1 pipe-and-hat format and HL7 FHIR Release 4 JSON resource bundles.

• Relational Persistence Tier: H2 database in PostgreSQL dialect (with seamless migration to PostgreSQL 16), governed by Flyway versioned migrations (V1, V2, V3) and database summary views.

12. Functional Requirements

CareConnect satisfies the following functional requirements:

• FR-01: User Authentication & Role Assignment — Authenticate users and enforce distinct capabilities for PATIENT, DOCTOR, and ADMIN roles.

• FR-02: Patient Registration & Profile Management — Capture legal demographics, contact details, emergency contacts, insurance policy numbers, and chronic illness history.

• FR-03: Clinical Charting & Specialty Templates — Support SOAP, Cardiology Consult, Pre-Op, and Discharge templates with separate Assessment and Treatment Plan inputs.

• FR-04: Note Immutability & Tamper Resistance — Finalized and signed clinical encounter notes cannot be edited; attempts return HTTP 409 Conflict.

• FR-05: CPOE Multi-Category Order Placement — Enable physicians to order laboratory tests (LOINC coded), diagnostic imaging (MRI, CT, X-Ray), and surgical/nursing procedures.

• FR-06: Electronic Prescribing & MAR — Prescribe medications with dosage, route, and frequency; record nurse administration events (GIVEN, HELD, REFUSED).

• FR-07: Drug-Drug Interaction Safety Checking — Screen proposed medications against active regimens and patient allergies, displaying contraindication severity banners.

• FR-08: Diagnostic Result Retrieval & Critical Alerts — View lab and imaging reports with reference ranges; automatically dispatch panic notifications when critical thresholds are crossed.

• FR-09: Interoperability Payload Generation — Format diagnostic results into HL7 v2.5.1 ORU^R01 messages and FHIR R4 DiagnosticReport resources.

• FR-10: Patient Portal Self-Service — View released health records, view prescriptions, schedule appointments, and exchange secure messages with assigned doctors.

13. Non-Functional Requirements

The platform adheres to stringent enterprise non-functional requirements:

• NFR-01: Security & Confidentiality — Stateless JWT authentication with HMAC-SHA384 tokens; BCrypt (cost factor 12) password hashing; zero plain-text credential storage.

• NFR-02: PHI Data Privacy & HIPAA Alignment — Access to patient health data is restricted strictly to authorized healthcare personnel and the authenticated patient.

• NFR-03: ACID Transactional Integrity — All multi-entity clinical workflows execute within atomic database transactions ensuring zero orphan orders or half-committed prescriptions.

• NFR-04: System Performance & Latency — Backend REST API response times maintain an average of < 80ms; frontend bundle builds into optimized minified chunks with sub-second page rendering.

• NFR-05: Determinism & Auditability — Every clinical mutation generates an immutable row in the audit_logs table detailing the actor, entity ID, action type, and modification timestamp.

14. Mandatory HCL Project Epics

The system fulfills all seven mandatory HCL Epics:

• Epic 1: Patient Registration and Demographics — Digital intake, insurance policy management, medical history collection, and verified patient search.

• Epic 2: Clinical Documentation — Multi-specialty templated documentation, assessment/plan separation, chronological retrieval, and tamper-resistant note locking.

• Epic 3: Computerized Provider Order Entry (CPOE) — Ordering for lab, imaging, and procedures with status tracking from dispatch to fulfillment.

• Epic 4: Medication Management — E-prescribing, reconciled patient drug lists, MAR administration tracking, and RxNorm-aligned DDI safety checks.

• Epic 5: Lab and Imaging Integration — Diagnostic report retrieval, abnormal range flags, automated critical panic alerts, and HL7/FHIR adapters.

• Epic 6: System Integration — Unified cross-module linkage connecting patients, encounters, orders, prescriptions, and results with comprehensive audit logging.

• Epic 7: Patient Portal — Self-service released record access, appointment booking, secure clinician messaging, and notification management.

15. Technology Stack Specification

Detailed technology stack breakdown:

• Frontend: React 18.3, Vite 5.4, React Router DOM 6.26, TanStack Query 5.56, Tailwind CSS 3.4, Lucide React Icons, Axios HTTP Client.

• Backend: Java 24, Spring Boot 3.5.7, Spring Security 6.4, Spring Data JPA / Hibernate 6.6, Flyway Migration 10.15, Spring AI 1.0.0-M6.

• Persistence: H2 Database 2.3 (PostgreSQL Compatibility Dialect), PostgreSQL 16 ready, HikariCP Connection Pool.

• Interoperability: HL7 v2.5.1 Pipe & Hat format, HL7 FHIR Release 4 JSON specification, LOINC & RxNorm standard coding.

• Automated Testing: JUnit 5, Spring Boot Test, Spring Security Test, MockMvc, AssertJ, Maven Surefire 3.5.4.

16. System Architecture & Data Flow

Data flows across the system follow strict architectural patterns:

1. Authentication: Client submits credentials to /api/auth/login; AuthService verifies BCrypt hash and issues signed JWT bearer token containing User ID and Role.

2. Clinical Encounter: Doctor initiates consultation in /doctor/patients/:id; selects template (SOAP/Cardiology); inputs findings; upon signing, backend locks the record and records an audit log.

3. CPOE Ordering: Doctor submits lab order via /api/orders/lab; order enters PLACED status; HospitalService generates a notification; technician records result via /api/results/lab; if value is critical, critical panic alert is immediately triggered.

4. E-Prescribing & MAR: Doctor prescribes medication; DrugInteractionService evaluates safety; upon commit, prescription becomes active; nurse logs MAR administration event via /api/patients/:id/mar.

5. Patient Interaction: Patient logs in; views released summaries; books appointments via /api/appointments; sends secure messages via /api/messages.

17. Database Design & Schema Specification

The CareConnect database incorporates 16 tables and 4 relational views:

• Core Tables: app_users, patient_profiles, doctors, departments, appointments, clinical_notes, medications_catalog, prescriptions, lab_orders, lab_results, imaging_orders, imaging_results, procedure_orders, medication_administrations, secure_messages, patient_notifications, audit_logs.

• Database Views:

- view_patient_summary: Aggregates patient demographics, primary physician, active prescription counts, latest encounter date, and pending lab counts.

- view_provider_schedule: Correlates provider schedules with appointments and patient references.

- view_medication_history: Joins prescriptions with completed MAR administration logs and adherence ratios.

- view_lab_orders_and_results: Correlates test orders with numerical values, reference ranges, and critical flags.

• Foreign Key Integrity: Foreign keys enforce parent-child integrity between users, patients, doctors, encounters, orders, results, and audit events.

18. Module Descriptions

CareConnect is structured into modular clinical subsystems:

• Registration & Demographics Module: Manages onboarding, demographic verification, and insurance policy details.

• Clinical Documentation Module: Handles multi-specialty progress notes with digital signature capture and immutability protection.

• CPOE Ordering Module: Centralizes ordering for laboratory diagnostic panels, diagnostic imaging modalities, and interventional procedures.

• Medication & MAR Module: Facilitates electronic prescribing, active drug reconciliation, MAR charting, and DDI contraindication checking.

• Diagnostic & Interoperability Module: Visualizes diagnostic reports, dispatches critical panic alerts, and serializes HL7/FHIR payloads.

• Patient Engagement Module: Provides patient portal self-service for appointment scheduling, record review, and secure messaging.

• Hospital Operations & Audit Module: Aggregates operational KPIs, bed occupancy, throughput analytics, and system-wide audit logging.

19. Implementation Methodology

The project adhered strictly to the Agile / Scrum delivery framework across four 2-week sprint iterations:

• Sprint 1: Project Foundation, Core Architecture, Flyway Migrations V1/V2, User Management & Patient Demographics (Epic 1).

• Sprint 2: Clinical Documentation Subsystem, Specialty Templates, Note Immutability & Tamper Resistance (Epic 2), CPOE Orders Engine (Epic 3).

• Sprint 3: Medication Management, Electronic Prescriptions, MAR Administration Logs & RxNorm DDI Pharmacology Safety Gateway (Epic 4).

• Sprint 4: Diagnostic Integration, Critical Panic Alerts, HL7/FHIR Gateway (Epic 5), Cross-Module Integration & Audit Trail (Epic 6), Patient Portal (Epic 7), Automated Verification.

20. Testing & Verification Strategy

CareConnect underwent rigorous automated testing across multiple testing levels:

• Unit & Domain Testing: Verified entity constraints, DDI contraindication logic, and HL7/FHIR message serialization.

• Service & Transactional Integration Testing: Executed in CareConnectEpicTests (8 tests) verifying multi-table ACID transactions, patient registration, note immutability locks (409 Conflict), CPOE life-cycles, MAR logs, and panic alerts.

• WebMvc Controller Testing: Executed in CareConnectWebMvcTests (7 tests) validating HTTP status codes, security filter chains, JSON payloads, and unauthorized access rejection.

• Frontend Production Verification: Vite production build transformed 2503 modules in 15.37 seconds with zero syntax errors.

• Total Automated Test Pass Rate: 15 of 15 tests PASSED (0 failures, 0 errors, 0 skipped).

21. Experimental & Verification Results

Actual verification results obtained during final execution:

• Test Execution Time: 30.067 seconds for complete backend Maven surefire suite.

• Clinical Immutability Gate: 100% of modification attempts on signed notes were blocked with HTTP 409 Conflict.

• DDI Safety Detection: High-severity contraindications (e.g., Lisinopril + Spironolactone) and allergy cross-reactions (Penicillin hypersensitivity) were flagged with 100% sensitivity.

• Critical Panic Alerting: Diagnostic results exceeding panic thresholds (Potassium > 6.2 mmol/L) triggered immediate physician notification events.

• Interoperability Conformance: Validated against HL7 v2.5.1 MSH/PID/OBR/OBX structure and HL7 FHIR Release 4 JSON specification.

22. System Limitations & Boundary Disclosures

In accordance with academic honesty and HCL guidelines, system boundaries are transparently documented:

• Pharmacology Knowledge Base: The DDI service utilizes a curated, standard NLM RxNorm interaction dataset. It is appropriately labelled as a demonstration gateway and does not substitute for commercial licensed compendia (e.g., First Databank or Wolters Kluwer Medi-Span).

• Diagnostic Interoperability: The HL7 and FHIR adapters generate syntactically valid standard payloads; however, they connect to an in-memory modular sandbox adapter rather than a live commercial hospital LIS/PACS connection.

• Non-Autonomous Clinical Advice: AI-assisted capabilities provide informational hospital SOP retrieval only and never autonomously prescribe medications or diagnose patients.

23. Future Scope & Enhancements

Future development phases for CareConnect may include:

• Integration with Commercial Pharmacology APIs: Connecting with live First Databank (FDB) or Lexicomp APIs for real-time commercial drug screening.

• Direct DICOM PACS Image Streaming: Embedding a zero-footprint CornerstoneJS DICOM viewer for real-time radiographic image streaming.

• Biometric & Smart Card Authentication: Implementing multi-factor WebAuthn and smart health card authentication for physician digital signatures.

• Mobile Native Applications: Developing companion iOS and Android patient engagement apps utilizing React Native.

24. Conclusion

CareConnect (Project Code: EHRS01) successfully demonstrates a robust, enterprise-grade Electronic Health Record system aligned with the official HCL Requirements Addendum. By integrating patient onboarding, clinical documentation, CPOE ordering, electronic prescribing with medication safety checks, diagnostic results with critical alerting, and patient self-service into an intuitive, responsive, and secure platform, the project achieves all required business objectives and functional acceptance criteria. Supported by passing automated test suites, clean production builds, and authentic documentation, CareConnect stands fully verified and ready for Academic Center of Excellence (COE) submission.

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